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Penetrator system for electrical submersible pumps

Abstract

The present disclosure relates to a wellhead penetrator system comprising a penetrator body comprising a hollowed cylindrical frame connecting a top end to a bottom end; a cable seal located within the penetrator body, comprising: a cylindrical core made of a polymer having three cylinder-shaped port holes configured to each provide a path for of the three insulated electrical wires; a top side, wherein the port holes extend upward from the top side of the cable seal; and a bottom side facing into the cavity located in the bottom end of the penetrator body, wherein the port holes extend downward from the bottom side of the cable seal; and a follower comprising a cylindrical metal body and cylindrical holes, wherein the follower seats against the bottom side of the cable seal so that port holes of the cable seal protrude through the cylindrical holes of the follower.

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Background/Summary

CROSS-REFERENCE SECTION (1) This application claims priority to U.S. Provisional Application No. 63/434,301, filed on Dec. 21, 2022, which is incorporated by reference herein in its entirety for all purposes.

FIELD OF THE DISCLOSURE
(1) The present disclosure relates, according to some embodiments, to wellhead penetrator systems for connecting downhole electric submersible pumps to electrical power sources.
BACKGROUND
(2) Electrical submersible pumps (ESPs) are used in the oil and gas industry to artificially lift fluids out of a wellbore. ESPs include downhole pumps and motors that connect to control systems at the wellhead via electrical cables. ESP systems often utilize penetrator systems to protect the cables (e.g., ESP cables, industrial cables, electrical power cables) at the wellhead so that power is continuously provided to the ESP motor. Because drilling operations continually occur at great depths, penetrator systems must be able to withstand increased amounts of pressure generated in the wellbore.
(3) Penetrator systems typically receive cables at a seal within the penetrator system, and at that point, the cables are cut (or spliced), and other electrical connections are made such that signals are provided to the control system at the wellhead. Penetrator systems that do provide for cables to pass through the penetrator system currently require inconvenient epoxy to be packed at the seal around the cable, and further are unable to withstand increased pressure.
(4) Improved penetrator systems allowing for passage of the cables above the wellhead are desired.
SUMMARY
(5) An electrical penetrator system according to the present disclosure includes a body having a

cable seal disposed therein. A plurality of electrical cables is disposed through the cable seal. A polymer insulator is coupled to the body above the cable seal, and the electrical cables extend in a direction uphole of the polymer insulator. The penetrator system further includes a plurality of compression seals disposed about the electrical cables at an upper end of the polymer insulator. A compression plate is coupled to the upper end of the polymer insulator and against the compression seals to thereby compress the compression seals about the electrical cables. A cavity is defined about the electrical cables below the cable seal. Notably, the cavity is devoid of any encapsulant, filling material, such as epoxy resin or putty.

(6) The present disclosure relates, according to some embodiments, to a wellhead penetrator system including a penetrator body including a cylindrical frame connecting a top end to a bottom end, the cylindrical frame having a hollowed interior configured to house other components of the wellhead penetrator system and an electrical submersible pumping (ESP) cable may include three electrical wires. The penetrator body may include the bottom end including a cavity configured to receive a cable that may be connected to an electrical submersible pump; and the top end configured to permit passage of the cable therethrough. The wellhead penetrator system may include a polymer insulator including a bottom face located within the penetrator body; a top face connected to the bottom face through an elongated central portion, the top face extending out of the top end of the penetrator body; and three cylinder-shaped cavities configured to each provide a path for one of the plurality of insulated electrical wires of the cable. The wellhead penetrator may include a compression plate seated on the top face of the polymer insulator and configured to fasten the wellhead penetrator system to a portion of the cable. The penetrator may include a cable seal including a cylindrical core made of a polymer having three cylinder-shaped port holes configured to each provide a path for of the three insulated electrical wires; a top side seated against the bottom face of the polymer insulator, wherein the port holes extend upward from the top side of the cable seal into the bottom face of the polymer insulator; and a bottom side facing into the cavity located in the bottom end of the penetrator body, wherein the port holes extend downward from the bottom side of the cable seal.

(7) According to some embodiments, the present disclosure relates to a wellhead penetrator system that may include a penetrator body including a cylindrical frame connecting a top end to a bottom end, the cylindrical frame having a hollowed interior configured to house other components of the penetrator system and an electrical submersible pumping (ESP) cable may include three electrical wires. The penetrator body may include the bottom end having a cavity configured to receive a cable that may be connected to an electrical submersible pump. The penetrator body may include the top end configured to permit the cable to pass therethrough. The wellhead penetrator system may include a cable seal located within the penetrator body, the cable seal including a cylindrical core made of a polymer having three cylinder-shaped port holes configured to each provide a path for the three insulated electrical wires; a top side, wherein the port holes extend upward from the top side of the cable seal; and a bottom side facing into the cavity located in the bottom end of the penetrator body, wherein the port holes extend downward from the bottom side of the cable seal. The wellhead penetrator system may include a follower may include a cylindrical metal body and three cylindrical holes, wherein the follower seats against the bottom side of the cable seal so that port holes of the cable seal protrude through the cylindrical holes of the follower.

(8) The wellhead penetrator may include a plurality of compression seals fitting into the cylinder-shaped cavities located at the top face of the polymer insulator where the polymer insulator meets the compression plate, wherein the compression plate may be configured to compress the plurality of compression seals about the cables at each site they pass through the polymer insulator, thereby forming a seal at each site.

(9) The wellhead penetrator may include a contoured cable positioner that may include a coupling portion may include an annular shape and an outside diameter that may be less than an inside diameter of the penetrator body, the coupling portion configured to seat inside the bottom end of

the penetrator body. The coupling portion may include an annular shape and an outside diameter that matches the outside diameter of the penetrator body where the components meet, wherein the outside diameter of the contoured portion narrows as it extends away from the penetrator body. In some embodiments, the inside diameters of each of the coupling portion and the contoured portion are large enough to receive the cable while wrapped in an armored sleeve.
(10) The compression seals may be made from a polymer may include a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone, and mixtures thereof. The polymer insulator may be made from a polymer may include a polyether ether ketone, a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone, and mixtures thereof. The compression plate may be made from a steel alloy, a polymer, or a mixture thereof. The compression plate may be from about 0.5 inches thick to about 4 inches thick. The cavity may be devoid any filling material.
(11) The wellhead penetrator system may include a follower may include a cylindrical metal body and three cylindrical holes, wherein the follower seats against the bottom side of the cable seal so that port holes of the cable seal protrude through the cylindrical holes of the follower. The cable seal may be one-piece and multi-point pressure activated. The port holes may extend downward from the bottom side of the cable seal are configured to seat against electrical wires of the cable having the lead jacket intact.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Some embodiments of the disclosure may be understood by referring, in part, to the present disclosure and the accompanying drawings, wherein:
(2) FIG. **1** illustrates a disclosed wellbore penetrator system connecting to a line box and a wellhead, according to an example embodiment of the disclosure;
(3) FIG. **2** illustrates a cross-sectional view of disclosed wellbore penetrator, according to an example embodiment of the disclosure;
(4) FIG. **3**A illustrates a top isometric view of a disclosed wellbore penetrator, according to an example embodiment of the disclosure;
(5) FIG. **3**B illustrates a side view of the disclosed wellbore penetrator of FIG. **3**A; according to an example embodiment of the disclosure;
(6) FIG. **3**C illustrates a side view of the disclosed wellbore penetrator of FIG. **3**A with body slid up to show the interior components, according to an example embodiment of the disclosure;
(7) FIG. **3**D illustrates a cross-sectional view of the wellbore penetrator of FIG. **3**A, according to an example embodiment of the disclosure; and
(8) FIG. **3**E illustrates a zoomed in cross-sectional view of the wellbore penetrator of FIG. **3**A, according to an example embodiment of the disclosure.
DETAILED DESCRIPTION
(9) The present disclosure relates to wellhead penetrator systems for creating a seal around cables, to prevent fluid and gas escape to the surface, as the cables pass through a wellhead. Cables may include, but are not limited to, electrical submersible pumping (ESP) cables, industrial cables, electrical power cables. Wellheads are a common intermediate point when connecting various components of an electric submersible pump system. For example, the wellhead penetrator systems disclosed herein may connect a down well electrical submersible pump to a line box and/or a control center through cables in a manner enables safe passage of the cable through the wellhead while providing a seal from wellbore pressures and atmosphere while not interrupting power

delivery as it is connected to the line box and/or control center.

(10) As shown in FIG. **1**, a disclosed wellhead penetrator system **100** may receive cables **106** from down well and connect them to a line box **104** upwell of the wellhead **102**. Cables **106** are generally multi-layered. At the core layer of the cables **106** are three metal conductor wires that each have a polymer insulation layer surrounding them. The polymer insulation layers are individually covered by lead jackets, which may or may not be individually covered by a polymer braid. The braid covered wires are grouped together by an armor layer that protects the cables **106**. As shown in FIG. **1**, the wellhead penetrator system **100** may receive the cables **106** that are fully covered by an armor layer **108**. The armor layer **108** may be removed as the cables **106** transition through the wellhead penetrator system **100** so that only a sub-layer (e.g., insulation layer) remains as the cables **106** exit the top of the wellhead penetrator system **100**. Disclosed wellhead penetrator systems **100**, as shown in FIG. **1**, do not require the cables **106** be spliced, which is a significant advantage over known penetrator systems. cables **106** may run directly through and then above disclosed wellhead penetrator systems **100** so they can be connected to a line box **104**.

(11) Known penetrator systems generally cut or splice cables at the seal portion of their penetrator systems. Disclosed wellhead penetrator systems **100** that do not require cutting or splicing of the cables provide enhanced sealing, stability, and pressure resistance in comparison to known systems, making them safer and more reliable. Component numbers increase incrementally based on which figure they are depicting. For example, component **100** in FIG. **1** will be annotated as component **200** in FIG. **2** and as component **300** in FIG. **3**. As shown in FIG. **2**, the cables **206** pass through the disclosed wellhead penetrator system **200** without being spliced. FIG. **2** illustrates a cross-sectional view of disclosed wellbore penetrator **200**.

(12) In some embodiments, as shown in FIG. **2**, the disclosed wellhead penetrator system **200** may include a penetrator body **210**, a polymer insulator **212**, a compression plate **214**, a cable seal **216**, and a follower **220**, in various permutations. For example, the wellhead penetrator system **200** may include the penetrator body **210**, the polymer insulator **212**, the compression plate **214**, and the cable seal **216**. In some embodiments, the wellhead penetrator system **200** may include the penetrator body **210**, the cable seal **216**, and the follower **220**.

(13) As shown in FIG. **2**, the disclosed wellhead penetrator system **200** may include the penetrator body **210** having a cylindrical frame connecting a top end to a bottom end. The cylindrical frame may have a hollowed interior configured to house other components of the wellhead penetrator system **200** along the cable **206**, if present. The bottom end of the penetrator body **210** may include a cavity configured to receive the cable **206** that is wrapped in the armored sleeve **208**. The cable **206**, as shown in FIG. **2**, may extend down from the bottom end of the penetrator body **210** (cable shown as **224** here) into a lower cavity **222** to connect to the electrical submersible pump. The cable **206** may pass through and out of the top end of the penetrator body **210** to connect to the line box (see FIG. **1**), or other electronic devices.

(14) In some embodiments, the penetrator body **210** may be made from a metal, a polymer, or a mixture thereof. The penetrator body **210** may be made from a metal including any steel alloys, such as zinc **4130**. The penetrator body **210** may be made from a polymer including a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone, and mixtures thereof. The penetrator body **210** may have an outside diameter ranging from about 1 inch to about 10 inches. For example, the penetrator body **210** may have an outside diameter of about 1 inch, or of about 2 inches, or of about 3 inches, or of about 4 inches, or of about 5 inches, or of about 6 inches, or of about 7 inches, or of about 8 inches, or of about 9 inches, or of about 10 inches, where about includes plus or minus 0.5 inches.

(15) The penetrator body **210** may form a cavity having an inside diameter ranging from about 0.5 inches to about 10 inches. For example, the penetrator body may have an inside diameter of about 0.5 inches, or about 1 inch, or of about 2 inches, or of about 3 inches, or of about 4 inches, or of

about 5 inches, or of about 6 inches, or of about 7 inches, or of about 8 inches, or of about 9 inches, or of about 10 inches, where about includes plus or minus 0.5 inches. The cavity of the penetrator body **210** may be devoid of any filling material when combined with the remaining components of the wellhead penetrator system **200**. Notably, the cavity may be devoid of filling material such as an epoxy resin or putty to be provided in the cavity defined around the cables where the cables enter the cable seal **216** from the downhole direction. The penetrator body **210** may have a length ranging from about 1 inch to about 50 inches, or more. For example, the penetrator body **210** may have a length of about 1 inch, or about 5 inches, or about 10 inches, or about 15 inches, or about 20 inches, or about 25 inches, or about 30 inches, or about 35 inches, or about 40 inches, or about 45 inches, or about 50 inches, where about includes plus or minus 2.5 inches.
(16) As shown in FIG. **2**, the disclosed wellhead penetrator system **200**, may include the polymer insulator **212**. The polymer insulator **212** may include a bottom face that is configured to seat inside the penetrator body **210** and even seat against the cable seal **216**. The polymer insulator **212** may include a top face connected to the bottom face through an elongated central portion. As shown in FIG. **2**, the top face may extend out of the top end of the penetrator body **210**. The polymer insulator **212** may include a plurality of cylinder-shaped cavities configured to each provide a path for one of the plurality of insulated wires of the cable **206**. The polymer insulator **212** may insulate the wellhead penetrator system **200** and cable **206** from electricity, heat, pressure and movement-based damage. The polymer insulator may also serve to provide a mechanical and pressure barrier interface with the wellhead for the purpose of retaining the penetrator system within the wellhead/wellhead hanger or other pressure control apparatus. This polymer insulator may be made from a metal including any steel alloys, such as zinc **4130** and may be coated or non-coated with insulating materials made from a polymer including a polyether ether ketone, a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone, and mixtures thereof.
(17) The polymer insulator **212** may be made from a polymer including a polyether ether ketone, a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone, and mixtures thereof. For example, the polymer insulator **212** may be a polyether ether ketone (PEEK) polymer. The polymer insulator **212** may have an outside diameter ranging from about 1 inch to about 10 inches. For example, the polymer insulator **212** may have an outside diameter of about 1 inch, or of about 2 inches, or of about 3 inches, or of about 4 inches, or of about 5 inches, or of about 6 inches, or of about 7 inches, or of about 8 inches, or of about 9 inches, or of about 10 inches, where about includes plus or minus 0.5 inches. The inside diameters of each of the cylinder-shaped cavities may include any diameter big enough to hold cables **206**, including ESP cables, industrial cables, or other electrical power cables.
(18) As shown in FIG. **2**, the disclosed wellhead penetrator system **200** may include a compression plate **214** seated on the top face of the polymer insulator **212**. The compression plate **214** may be configured to fasten the wellhead penetrator system **200** to a portion of the cable **206** so that movement caused from oil production activities does not loosen any of the seals or displace any component configurations. The compression plate **214** may stick to the polymer insulator **212** through the action of compression, an adhesive, or even threadably screwing onto other components of the wellhead penetrator system **200**. The compression plate **214** may have an outside diameter ranging from about 1 inch to about 10 inches. For example, the compression plate **214** may have an outside diameter of about 1 inch, or of about 2 inches, or of about 3 inches, or of about 4 inches, or of about 5 inches, or of about 6 inches, or of about 7 inches, or of about 8 inches, or of about 9 inches, or of about 10 inches, where about includes plus or minus 0.5 inches. The compression plate **214** may have a thickness ranging from about 0.5 inches to about 4 inches. For example, the compression plate **214** may have a thickness of about 0.5 inches, or about 1 inch, or

about 2 inches, or about 3 inches, or about 4 inches, where about includes plus or minus 0.5 inches. The compression plate **214** may be made of any metal (e.g., steel), polymer, or a mixture thereof (e.g., polymer coated metal).

(19) FIG. **2** discloses the wellhead penetrator system **200** including the cable seal **216**. The cable seal **216** may be a multi-point pressure activated one-piece cable seal **216**. Disclosed cable seal **216** may have pressure-activated sealing capabilities at pressures up to and beyond 5,000 psi. For example, the cable seal **216** may provide for a pressure activated sealing at a pressure of about 1,000 psi, or about 2,000 psi, or about 3,000 psi, or about 4,000 psi, or about 5,000 psi, or about 6,000 psi, or about 7,000 psi, or about 8,000 psi, or about 9,000 psi, or about 10,000 psi, 11,000 psi, 12,000 psi, 13,000 psi, 14,000 psi, 15,000 psi, 16,000 psi, 17,000 psi, 18,000 psi, 19,000 psi, 20,000 psi where about includes plus or minus 500 psi. The cable seal **216** may include a cylindrical core made of a polymer having multiple (e.g., three) cylinder-shaped port holes configured to each provide a path for the insulated electrical wires of the cable **206**. The cable seal **216** may include a top side that may be seated against the bottom face of the polymer insulator **212**, wherein the port holes extend upward from the top side of the cable seal **216** into the bottom face of the polymer insulator **212**. The cable seal **216** may include a bottom side facing into the cavity located in the bottom end of the penetrator body **210** so that the port holes extend downward from the bottom side of the cable seal **216**. In some embodiments, the port holes extend downward from the bottom side of the cable seal **216** and receive the cables **206** that have had the armored sleeve **208** removed. Where the cables **206** meet the port holes, the cables **206** may have the armored sleeve **208** and braid removed so the lead jacket remains intact.

(20) As shown in FIG. **2**, the cable seal **216** may be held in place via a contoured cable positioner **226**. The cable seal **216** may be designed to withstand downhole well pressure. The cable seal **216** may include a first set of projections that extends towards a polymer insulator **212** and a second set of projections that extend in an opposite direction, towards the cavity of the penetrator body **210**. The projections may define cylindrical openings for receiving the cables **206** therethrough. In some embodiments, the cable seal **216** may be formed of a polymer (e.g., rubber) to facilitate a seal (e.g., mechanical seal) with the body.

(21) The cable seal **216** may include an outside diameter ranging from about 1 inch to about 10 inches. For example, the cable seal **216** may have an outside diameter of about 1 inch, or of about 2 inches, or of about 3 inches, or of about 4 inches, or of about 5 inches, or of about 6 inches, or of about 7 inches, or of about 8 inches, or of about 9 inches, or of about 10 inches, where about includes plus or minus 0.5 inches. The inside diameters of each of the port holes may include any diameter big enough to hold any known cables **206**. The cable seal **216** may be made from a polymer including a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone, and mixtures thereof.

(22) As shown in FIG. **2**, in some embodiments, the disclosed wellhead penetrator system **200** may include the compression seal **218** that fits into a cylinder-shaped cavity located at the top face of the polymer insulator **212** where the polymer insulator **212** meets the compression plate **214**. The compression plate **214** may be configured to compress the compression seals **218** about the cables **206** at each site they pass through the polymer insulator **212**, thereby forming a seal at each site. The wellhead penetrator system **200** may include the plurality of compression seals **218**, such as ranging from 1-5 compression seals **218**. The compression seal **218** may be made from a polymer including a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone, and mixtures thereof. The compression seals **218** may be generally cylindrical in shape and may have an inside diameter large enough for the cables **206** to fit through and may have an outside diameter large enough to seal a portion of each cylinder-shaped cavity of the polymer insulator **212**.

(23) In some embodiments, the cable seal **216** may abut against the polymer insulator **212**. In some embodiments, the polymer insulator **212** may be formed of PEEK material that insulates the cables **206** passing through the polymer insulator **212**. Each cable **206** may also have rubber or polymer insulator wrapped around the cables. Each of the cables **206** may have a corresponding compression seal **218** disposed about the cable **206** at an upper portion of the polymer insulator **212**. The compression plate **214** may be utilized to bear down on the compression seals **218** to form a seal about the cables **206** where the cables **206** exit the polymer insulator **212**. In some embodiments, the compression plate **214** may be threaded to the polymer insulator **212** via set screws, thus applying the desired compression force to form the compression seals **218**.

(24) In some embodiments, as shown in FIG. **2**, the wellhead penetrator system **200** may include the follower **220** including a cylindrical metal body and from 1-3 cylindrical holes. The follower **220** may be configured to seat against the bottom side of the cable seal **216** so that the port holes of the cable seal protrude through the cylindrical holes of the follower **220**. The follower **220** may have an outside diameter of about 1 inch, or of about 2 inches, or of about 3 inches, or of about 4 inches, or of about 5 inches, or of about 6 inches, or of about 7 inches, or of about 8 inches, or of about 9 inches, or of about 10 inches, where about includes plus or minus 0.5 inches. The outside diameter of the follower **220** may contract part way through the length of the follower **220**. For example, the follower **220** may have an outside diameter of about 4 inches towards its top and have an outside diameter of about 3 inches towards its bottom. The difference in outside diameter between the top portion and bottom portion of the follower **220** may from about 0.1 inches to about 2 inches. For example, the difference in outside diameter between the top portion and bottom portion of the follower **220** may be about 0.1 inches, or about 0.25 inches, or about 0.5 inches, or about 0.75 inches, or about 1 inch, or about 1.25 inches, or about 1.5 inches, or about 1.75 inches, or about 2 inches, where about includes 0.125 inches. The inside diameters of each of the port holes may include any diameter big enough to hold any cables **206**, including ESP cables, industrial cables, or other electrical power cables. The follower **220** may be made of any known metal, including steel alloys.

(25) As shown in FIG. **2**, the wellhead penetrator system **200** may include the contoured cable positioner **226** configured to couple to the bottom end of the penetrator body **210**. The contoured cable positioner **226** may include a coupling portion and a contoured portion. The coupling portion may have an annular shape and an outside diameter that is less than the inside diameter of the penetrator body **210**. The coupling portion may be configured to seat inside the bottom end of the penetrator body **210**. The contoured cable positioner **226** may be coupled to the penetrator body **210** via a fastener, such as a screw or the like. The contoured portion may have an annular shape and an outside diameter that matches the outside diameter of the penetrator body **210** so that the surfaces are substantially smooth when coupled where the components meet. The outside diameter of the contoured portion may narrow as it extends away from the penetrator body **210**. In some embodiments, the inside diameters of each of the coupling portion and the contoured portion are large enough to receive the cable **206** while wrapped in an armored sleeve **208**.

(26) According to some embodiments, the portion of the cables **206** that exit the armored sleeve **208** and enter the cable seal **216** may be housed in a lead jacket to protect against decompression and any adverse well or well fluid conditions. The cavity may be defined in the region of the penetrator body **210** where the cables **206** exit the armored sleeve **208** and enter the cable seal **216**. In prior art arrangements, a filler material, such as an epoxy resin, is required to be disposed within a similar cavity in order to protect against downhole well pressure. However, according to the principles of the present disclosure, no such filler material is required to be disposed in the cavity as the integrity of the cable seal **216** and the upper compression seals **218** are sufficient to guard against undesirable downhole well pressure. Disclosed configurations advantageously reduce material needed for manufacture of the penetrator system as well as improves ease of use. This configuration also allows the cable seal **216** to expand and contract dynamically thus creating a

pressure balanced sealing mechanism. The allowance of movement due to lack of filler material prevents the cable seal **216** from mechanical damage during expansion and contraction thus improving longevity of the sealing system.

(27) FIGS. **3**A-**3**E illustrate the disclosed wellhead penetrator system **300** having the above-described components and engaged with a cable **306** having an armored sleeve **308**. The penetrator system **300** may include the cable seal **316** (i.e., boot seal) for receiving cables **306** from downhole and passing the cables **306** through the polymer insulator **312** (such as one made of polyether ether ketone or “PEEK” material) and uphole to the wellhead. The cable seal **316** may be housed within the penetrator body **310**, such as a steel housing, that also retains the PEEK insulator and armored sleeve **308** that encapsulates the cables **306** in a downhole direction. The wellhead penetrator system **300** may further includes the contoured cable positioner **326** where the cables **306** exit the armored sleeve **308** and enter the wellhead penetrator system **300**. The wellhead penetrator system **300** may also include compression seals (see **218** from FIG. **2**) provided at the uphole end of the PEEK polymer insulator **312**. The compression plate **314** may be provided to compress the compression seals around the cables **308** to further protect against downhole pressure. The wellhead penetrator system **300** described herein may not need an encapsulant filling material such as epoxy resin or putty to be provided in the cavity defined by the penetrator body **310** and around the cables **306** where the cables **306** enter the cable seal **316** from the downhole direction. Further, the wellhead penetrator system **300** may be rated for high pressure, as much as 5,000 psi and more. For example, the wellhead penetrator system **300** may be rated for a pressure of about 1,000 psi, or about 2,000 psi, or about 3,000 psi, or about 4,000 psi, or about 5,000 psi, or about 6,000 psi, or about 7,000 psi, or about 8,000 psi, or about 9,000 psi, or about 10,000 psi, or about 11,000 psi, or about 12,000 psi, or about 13,000 psi, or about 14,000 psi, or about 15,000 psi, or about 16,000 psi, or about 17,000 psi, or about 18,000 psi, or about 19,000 psi, or about 20,000 psi, where about includes plus or minus 500 psi. The wellhead penetrator system **300** may further protect against harmful elements, such as fluid intrusion, gas intrusion, debris, or the like. As shown in FIGS. **3**A and **3**B, disclosed wellhead penetrator system **300** may be assembled so that the polymer insulator **312** protrudes outward above the penetrator body **310**. Above the polymer insulator **312** may be compression plate **314**, and the cable **306** may extend outward from the compression plate **314** so that it may connect to above well components, such as the line box.

(28) FIG. **3**C illustrates the wellhead penetrator system **300** of FIG. **3**A, but with the penetrator body **310** slid up so that the components of this embodiment are viewable. As shown in FIG. **3**C, the wellhead penetrator system **300** may include the contoured cable positioner **326** receiving the cable **306** wrapped in armor **308**. Follower **320** may seat above the contoured cable positioner **326** with the cable seal **316** seating above the follower **320**.

(29) FIGS. **3**D and **3**E are cross-sectional views of the wellhead penetrator system **300** of FIG. **3**A. Each of these figures show the connectivity of an embodiment where the compression plate **314** sits above the polymer insulator **312**, which is above the cable seal **316** and the follower **320**. As shown in FIG. **3**D, the port holes of the cable seal **316** may protrude through the cylindrical holes of the follower **320** while extending into the cavity formed by the penetrator body **310**. Below the penetrator body is the contoured cable positioner **326**. FIGS. **3**D and **3**E show how an un-spliced cable **306** may travel through the entire wellhead penetrator system **300**.

(30) The above descriptions of the implementations of the present disclosure have been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present disclosure to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the present disclosure be limited not by this detailed description, but rather by the claims of this application. As will be understood by those familiar with the art, the present disclosure may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. Accordingly, the present disclosure is intended to be illustrative, but not limiting, of the scope of the present disclosure, which is set forth

in the following claims.
(31) The figures and descriptions provided herein may have been simplified to illustrate aspects that are relevant for a clear understanding of the herein described devices, systems, and methods, while eliminating, for the purpose of clarity, other aspects that may be found in typical similar devices, systems, and methods. Those of ordinary skill may recognize that other elements and/or operations may be desirable and/or necessary to implement the devices, systems, and methods described herein. But because such elements and operations are well known in the art, and because they do not facilitate a better understanding of the present disclosure, a discussion of such elements and operations may not be provided herein. However, the present disclosure is deemed to inherently include all such elements, variations, and modifications to the described aspects that would be known to those of ordinary skill in the art.
(32) The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. For example, as used herein, the singular forms “a”, “an” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.
(33) Although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. That is, terms such as “first,” “second,” and other numerical terms, when used herein, do not imply a sequence or order unless clearly indicated by the context.
(34) Reference in the specification to “one implementation” or “an implementation” means that a particular feature, structure, or characteristic described in connection with the implementation is included in at least one implementation of the disclosure. The appearances of the phrase “in one implementation,” “in some implementations,” “in one instance,” “in some instances,” “in one case,” “in some cases,” “in one embodiment,” or “in some embodiments” in various places in the specification are not necessarily all referring to the same implementation or embodiment.

Claims

1. A wellhead penetrator system comprising: (a) a penetrator body comprising: (i) a cylindrical frame connecting a top end to a bottom end, the cylindrical frame having a hollowed interior configured to house other components of the wellhead penetrator system and a cable comprising three electrical wires; (ii) the bottom end comprising a cavity configured to receive the cable that is connected to an electrical submersible pump; and (iii) the top end configured to permit passage of the cable therethrough; (b) a polymer insulator comprising: (i) a bottom face located within the penetrator body; (ii) a top face connected to the bottom face through an elongated central portion, the top face extending out of the top end of the penetrator body; and (iii) three cylinder-shaped cavities configured to each provide a path for one of the plurality of insulated electrical wires of the cable; (c) a compression plate seated on the top face of the polymer insulator and configured to fasten the wellhead penetrator system to a portion of the cable; and (d) a cable seal, comprising: (i) a polymer having three cylinder-shaped port holes configured to each provide a path for of the three insulated electrical wires; (ii) a top side seated against the bottom face of the polymer insulator, wherein the port holes extend upward from the top side of the cable seal into the bottom

face of the polymer insulator; and (iii) a bottom side facing into the cavity located in the bottom end of the penetrator body, wherein the port holes extend downward from the bottom side of the cable seal.

2. The wellhead penetrator system according to claim 1, further comprising: a plurality of compression seals fitting into the cylinder-shaped cavities located at the top face of the polymer insulator where the polymer insulator meets the compression plate, wherein the compression plate is configured to compress the plurality of compression seals about the cables at each site they pass through the polymer insulator, thereby forming a seal at each site.

3. The wellhead penetrator system according to claim 2, wherein the compression seals are made from a polymer comprising one or more of a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, and a silicone.

4. The wellhead penetrator system according to claim 1, further comprising: a contoured cable positioner comprising: a coupling portion comprising an annular shape and an outside diameter that is less than an inside diameter of the penetrator body, the coupling portion configured to seat inside the bottom end of the penetrator body; and a contoured portion comprising an annular shape and an outside diameter that matches the outside diameter of the penetrator body where the components meet, wherein the outside diameter of the contoured portion narrows as it extends away from the penetrator body; wherein the inside diameters of each of the coupling portion and the contoured portion are large enough to receive the cable while wrapped in an armored sleeve.

5. The wellhead penetrator system according to claim 1, wherein the polymer insulator is made from one or more of a polymer comprising a polyether ether ketone, a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone.

6. The wellhead penetrator system according to claim 1, wherein the compression plate is made from a steel alloy, a polymer, or a mixture thereof.

7. The wellhead penetrator system according to claim 1, wherein the compression plate is from about 0.5 inches thick to about 4 inches thick.

8. The wellhead penetrator system according to claim 1, wherein the cavity is devoid any filling material.

9. The wellhead penetrator system according to claim 1, further comprising: a follower comprising a cylindrical metal body and three cylindrical holes, wherein the follower seats against the bottom side of the cable seal so that port holes of the cable seal protrude through the cylindrical holes of the follower.

10. The wellhead penetrator system according to claim 1, wherein the cable seal is one-piece and multi-point pressure activated.

11. The wellhead penetrator system according to claim 1, wherein the port holes extending downward from the bottom side of the cable seal are configured to seat against the electrical wires of the cable having a lead jacket intact.

12. A wellhead penetrator system comprising: (a) a penetrator body comprising: (i) a cylindrical frame connecting a top end to a bottom end, the cylindrical frame having a hollowed interior configured to house other components of the penetrator system and a cable comprising three electrical wires; (ii) the bottom end comprising a cavity configured to receive the cable that is connected to an electrical submersible pump; and (iii) the top end configured to permit the cable to pass therethrough; (b) a cable seal located within the penetrator body, comprising: (i) a polymer having three cylinder-shaped port holes configured to each provide a path for the three insulated electrical wires; (ii) a top side, wherein the port holes extend upward from the top side of the cable seal; and (iii) a bottom side facing into the cavity located in the bottom end of the penetrator body, wherein the port holes extend downward from the bottom side of the cable seal; and (c) a follower comprising a cylindrical metal body and three cylindrical holes, wherein the follower seats against

the bottom side of the cable seal so that port holes of the cable seal protrude through the cylindrical holes of the follower.

13. The wellhead penetrator system according to claim 12, further comprising: a polymer insulator comprising: (i) a bottom face located within the penetrator body and fitting up against the top side of the cable seal; (ii) a top face connected to the bottom face through an elongated central portion, the top face extending out of the top end of the penetrator body; and (iii) three cylinder-shaped cavities configured to each provide a path for one of the plurality of insulated electrical wires of the cable.

14. The wellhead penetrator system according to claim 13, further comprising: a compression plate seated on the top face of the polymer insulator and configured to fasten the wellhead penetrator system to a portion of the cable.

15. The wellhead penetrator system according to claim 14, comprising at least one of: the compression plate is made from a steel alloy, a polymer, or a mixture thereof, and the compression plate is from about 0.5 inches thick to about 4 inches thick.

16. The wellhead penetrator system according to claim 14, further comprising: a plurality of compression seals fitting into the cylinder-shaped cavities located at the top face of the polymer insulator where the polymer insulator meets the compression plate, wherein the compression plate is configured to compress the plurality of compression seals about the cables at each site they pass through the polymer insulator, thereby forming a seal at each site.

17. The wellhead penetrator system according to claim 16, wherein the compression seals are made from a polymer comprising one or more of polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone.

18. The wellhead penetrator system according to claim 13, wherein the polymer insulator is made from one or more of a polymer comprising a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone.

19. The wellhead penetrator system according to claim 12, wherein the cable seal is made from one or more of a polymer comprising a polyether ether ketone, a polyethylene, a polypropylene, a polystyrene, a polyvinyl chloride, a synthetic rubber, a phenol formaldehyde resin, a neoprene, a nylon, a polyacrylonitrile, a polyvinyl butyral, a silicone.

20. The wellhead penetrator system according to claim 12, wherein the penetrator body is made from a metal, a polymer, or a mixture thereof.
